

Saleh Eddine Khalfaoui

Software Engineer | Backend & Systems Engineering | AI Applications

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Software Engineer based in Tunisia, focused on backend engineering, system design, and distributed architecture. Builds Java/Spring Boot microservices with Apache Kafka, REST APIs, and Retrieval-Augmented Generation (RAG), plus full-stack work, databases, and Docker. Final-year student at ESPRIT, available for a 2027 end-of-studies internship (PFE).

Professional Experience

Software Engineering Intern — Amen Bank, Tunis, Tunisia

June 2026 – August 2026

- Developed **Java/Spring Boot** microservices and REST APIs for an AI-powered banking application.
- Integrated **Apache Kafka** for asynchronous, event-driven communication between services.
- Integrated **Retrieval-Augmented Generation (RAG)** and large language models into the backend.
- Containerized the services with **Docker** and Git, keeping boundaries decoupled and maintainable.

Web & BI Developer — Mobelite Labs, Tunis, Tunisia

February 2024 – January 2025

- Developed responsive **React** and **Next.js** interfaces integrated with REST APIs.
- Integrated **5 Power BI** dashboards with those APIs across **10K+** records.
- Automated **ETL** workflows and reduced manual reporting time by **50%**.
- Improved decision-making efficiency by **30%** through centralized reporting, in Agile sprints with Jira.

Software Engineering Intern — Amen Bank, Tunis, Tunisia

Summer 2025

- Developed web and backend features, including REST APIs and database-backed application logic.
- Implemented and tested features from requirements for reliability and code quality.

Projects

ByteBattle — Competitive programming platform with AI-assisted features

- Built challenge workflows, submission scoring, and leaderboards with React, NestJS, and MongoDB.
- Added JWT authentication, role-based access control, and OpenAI API integration in NestJS.
- Containerized the services with Docker for reproducible, deployment-ready environments.

E-commerce microservices — Separate Spring Boot services with independent databases

- Designed Spring Boot microservices with separate PostgreSQL databases and REST APIs.
- Implemented a Spring Cloud API gateway and containerized the stack with Docker.

NutriTrace — Monolithic Laravel MVC application for traceability

- Developed a Laravel 12 MVC application in PHP for traceability, with MySQL models and Leaflet maps.
- Added authentication and role-based access control with Spatie Permission on a REST monolith.

CargoLink — Intelligent freight platform with map-based routing and RAG

- Built a Django MVT application in Python with templates, the ORM, and HTML/CSS/JavaScript.
- Integrated Leaflet and the OpenRouteService API for route, distance, and ETA.
- Integrated Retrieval-Augmented Generation (RAG) for context-aware assistance grounded in logistics data.

Technical Skills

Languages — Java, Python, JavaScript, TypeScript, PHP, SQL

Backend — Spring Boot, NestJS, Node.js, Django, Laravel, Spring Cloud, REST APIs

Architecture — Microservices, Apache Kafka, event-driven architecture, distributed systems, system design, MVC

AI — Retrieval-Augmented Generation (RAG), LLMs, AI-powered applications, OpenAI API

Frontend — React, Next.js, HTML, CSS

Databases — PostgreSQL, MySQL, MongoDB

DevOps — Docker, Git, GitHub, Linux, Jira, Agile/Scrum

Data & BI — ETL, Power BI

Education

Engineering Cycle in Computer Science (Web & Software Engineering)

ESPRIT, Tunis

2024 – Present (expected 2027)

Bachelor's Degree in Software Engineering and Information Systems

École Polytechnique Internationale Privée de Tunis

2021 – 2024

Languages — English: Fluent French: Fluent Arabic: Native